

Quantum Complexity

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By the end of our 2 lectures you should be able to:

- Name several classical/quantum complexity classes.
- Give an example of a NP-complete and QMA-complete problem.
- Describe the Circuit to local-Hamiltonian reduction.
- Explain the connection between non-local games and interactive proofs.

Outline

- ① Basics of (Quantum) Complexity Theory.
- ② k -local Hamiltonian problem is QMA-complete.
- ③ Interactive proofs and other complexity classes.

Basics of Quantum Complexity Theory

What is complexity theory?

Question: What are some examples of “resources” which are related to solving computational problems?

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Complexity class	Resources / features
P	time
PSPACE	memory
BPP	time + randomness
BQP	quantum computation
NP, MA	classical proofs / randomness + proofs
QMA	quantum proofs
IP, MIP	interactive proofs
MIP*	interaction + entanglement
QMA(2)	unentangled quantum proofs

What is a computational problem ?

Is $1 + 1 = 2$?

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Given integers x, y , **and** z , **does** $x + y = z$.

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Given integers x, y , **and** z , **does** $x + y = z$.

Given integer n **is** n **a prime ?**

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Given a computer program A , does A halt?

What is a computational problem ?

$\{0, 1\}^* = \{\epsilon, 0, 1, 00, 01, 10, 11, 000, \dots\}$ denotes set of all finite strings.

A *language* or decision problem is a subset $L \subseteq \{0, 1\}^*$.

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A *language* or decision problem is a subset $L \subseteq \{0, 1\}^*$.

Strings 01110111 can encode mathematical objects.

$L = \{\textbf{input:}$ mathematical object x : $\textbf{statement:}$ $s(x)$ is true $\}$.

Equivalently, a language is a yes/no problem:

$x \in L \iff \text{YES,} \quad x \notin L \iff \text{NO.}$

e.g: $L := \{n \in \mathbb{N} : n \text{ is prime}\} \iff \text{Is a given integer } n \text{ is prime?}$

Examples

$$L_{\text{even}} = \{\text{strings } x \in \{0,1\}^* : x \text{ has an even number of 1's}\},$$

$$L_{\text{prime}} = \{n \in \mathbb{N} : n \text{ is prime}\},$$

$$L_{3\text{-col}} = \{\text{graphs } G : G \text{ is a 3 colourable graph}\},$$

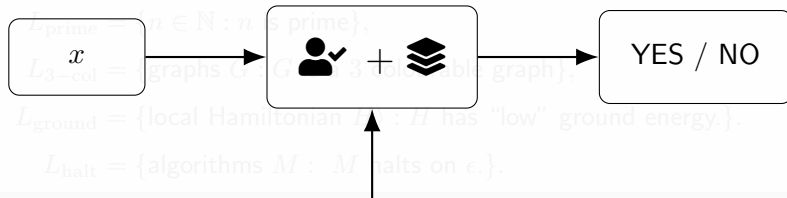
$$L_{\text{ground}} = \{\text{local Hamiltonian } H : H \text{ has "low" ground energy.}\}.$$

$$L_{\text{halt}} = \{\text{algorithms } M : M \text{ halts on } \epsilon.\}.$$

Here we assume the object is encoded as a string.

Examples

Complexity theory studies the resources needed to classify an input as YES or NO.



Here we assume

 = time, memory, randomness, proof y ,
quantum gates (quantum time), quantum proof $|\psi\rangle$...

How do we model a computer?

Classical computation: many equivalent models

Extended Church–Turing thesis (informal)

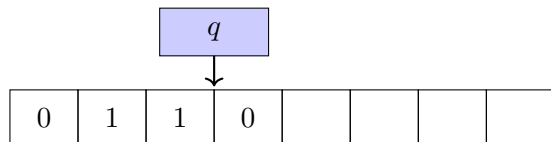
Any “reasonable” model of computation can be efficiently simulated by a Turing machine.

Classical computation: many equivalent models

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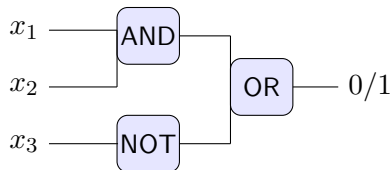
Any “reasonable” model of computation can be efficiently simulated by a Turing machine.

Turing machine



Reads and writes symbols
on a tape step-by-step

Boolean circuit



A boolean circuit determines
 $C : \{0, 1\}^n \rightarrow \{0, 1\}$

Turing machines $\approx$ Boolean circuits $\approx$ standard programming languages.

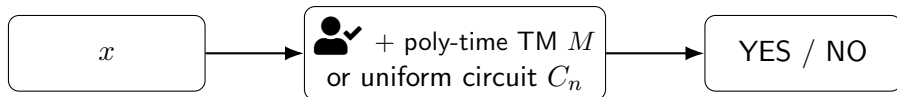
Time Complexity (The class P)

Definition of P

A language L is in P if there is a polynomial-time algorithm deciding if $x \in L$.

Equivalently, there is a **uniform** family of Boolean circuits $\{C_n\}$ such that

$$C_n(x) = 1 \iff x \in L \quad \text{for all } x \in \{0, 1\}^n.$$



Polynomial time = “efficient”.

What are some examples of problems that can be solved efficiently?

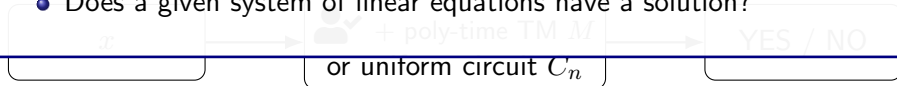
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- Does a string $x \in \{0,1\}^*$ contain an even number of 1s?
- Is a given integer n prime?
- Does a given system of linear equations have a solution?



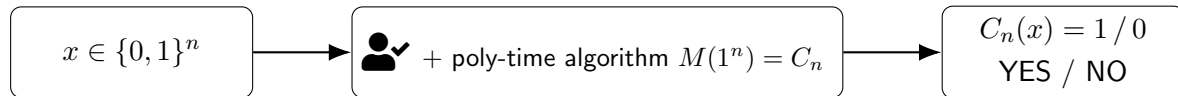
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Uniform circuits (Informal)

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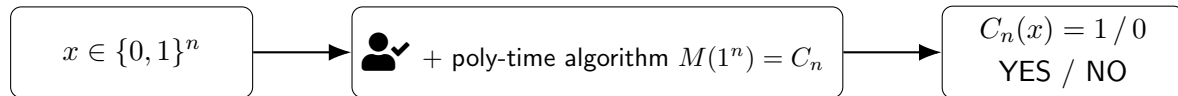
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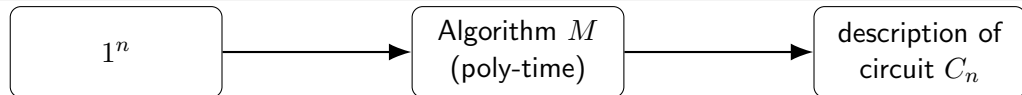
$$C_2 = x_1 \oplus x_2, \quad C_3 = x_1 \oplus x_2 \oplus x_3, \quad \dots \quad C_n = x_1 \oplus \dots \oplus x_n.$$

$$M(1^n) = C_n$$

Uniform Circuits

Uniformity

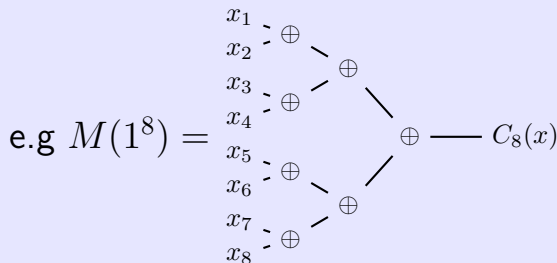
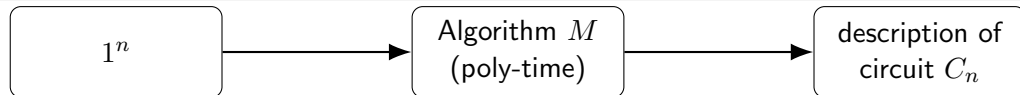
A family of Boolean circuits $\{C_n\}_{n \geq 1}$ is *uniform* if there exists a polynomial p and an algorithm M which, on input 1^n , outputs a description of the circuit C_n in at most $p(n)$ steps.



Uniform Circuits

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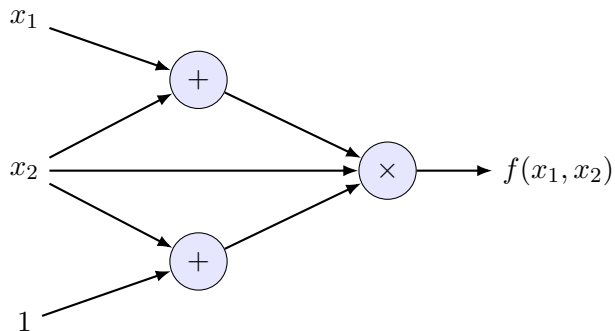
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Time + Other Resources.

Arithmetic Circuits

An arithmetic circuit A computes a polynomial in $\mathbb{F}[x_1, \dots, x_n]$ using gates $+$, $\times$.



$$f(x_1, x_2) = (x_1 + x_2)x_2(x_2 + 1).$$

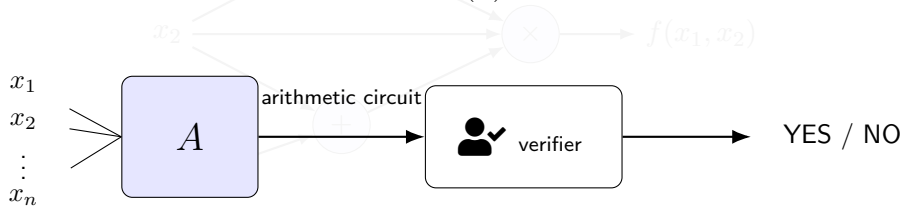
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Polynomial Identity Testing:

$L_{IT} = \{\text{arithmetic circuits } A : A(x_1, \dots, x_n) \equiv 0\}$.

Given an arithmetic circuit A decide if $A(x) = 0$ for all x .

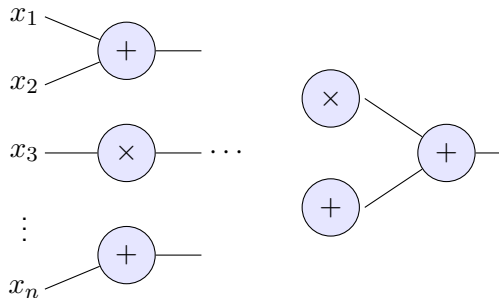


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Arithmetic circuits are succinct

We can't just expand the polynomial.

Arithmetic circuit



Expanded polynomial

$$\begin{aligned} &1 + x_1 + x_2 + \cdots + x_n \\ &+ x_1x_2 + x_1x_3 + \cdots + x_{n-1}x_n \\ &+ x_1x_2x_3 + \cdots \\ &\vdots \\ &+ x_1x_2 \cdots x_n. \end{aligned}$$

An arithmetic circuit of size n can represent a polynomial with 2^n monomials.

We can evaluate the circuit $A(x)$ efficiently.

To decide if $A(x) = 0$ for all x do we have to check all points $x = (x_1, \dots, x_n)$?

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Suppose $f(x)$ is a non-zero polynomial of degree d . How many zeros can it have?

Randomness as a resources

To decide if $A(x) = 0$ for all x do we have to check all points $x = (x_1, \dots, x_n)$?

Suppose $f(x)$ is a non-zero polynomial of degree d . How many zeros can it have?

Choose a uniformly random point $x = (x_1, \dots, x_n) \in \mathbb{F}^n$ and evaluate $A(x)$.

YES instance

If $A \equiv 0$, then $A(x) = 0$ for every x .

NO instance

If $\deg(A) \leq d$ and $A \not\equiv 0$, then
Schwartz–Zippel gives

$$\Pr_{x \in \mathbb{F}^n} [A(x) = 0] \leq \frac{d}{|\mathbb{F}|}.$$

Thus a few random evaluations give a highly reliable test.

Randomness as a resources

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Open question: Is randomness really necessary here?

Schwartz–Zippel gives

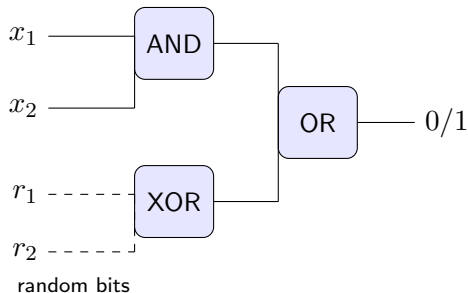
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Adding randomness: BPP

A randomized circuit is a Boolean circuit with access to uniformly random bits.

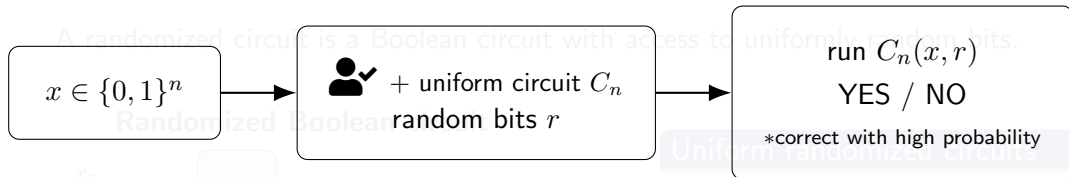
Randomized Boolean circuit



Uniform randomized circuits

A uniform randomized circuit family is a uniform family of Boolean circuits $\{C_n(x, r)\}$, where r denotes uniformly random bits.

Definition of BPP (informal)



Problems Solvable on a Quantum Computer

Two computational challenges

Find x such that

$$2^x \equiv 13 \pmod{17}.$$

Find primes p and q such that

$$pq = 221.$$

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Can you find x such that

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Can you factor

$$1073?$$

Does randomness seem helpful?

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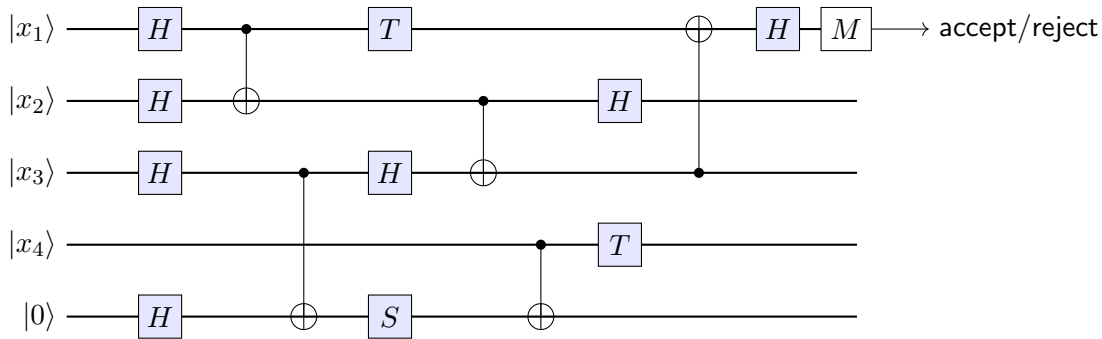
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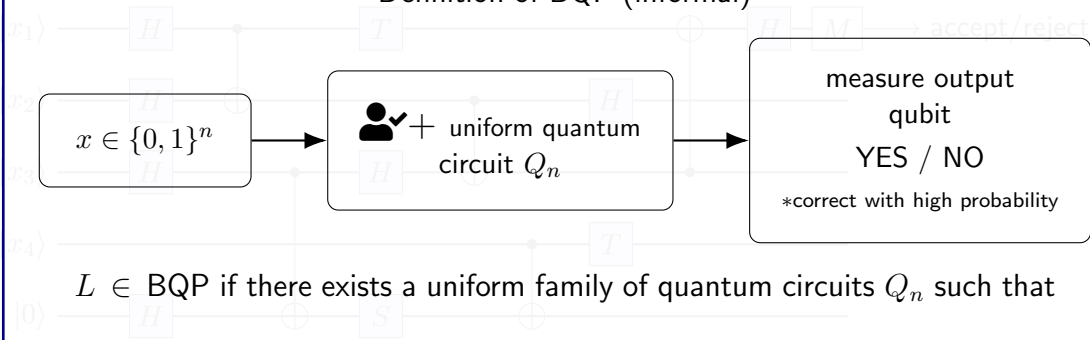
Both problems can be solved efficiently using Shor's quantum algorithm.



Uniform quantum circuits

A family of quantum circuits $\{Q_n\}_{n \geq 1}$ is *uniform* if there exists a polynomial-time algorithm which, on input 1^n , outputs a description of Q_n .

Definition of BQP (informal)



$L \in \text{BQP}$ if there exists a uniform family of quantum circuits Q_n such that

$$x \in L \implies \Pr[Q_n(x) = 1] \geq \frac{2}{3} \quad (\text{completeness})$$

A family of quantum circuits $\{Q_n\}_{n \geq 1}$ is *uniform* if there exists a polynomial-time algorithm which, on input 1^n , outputs a description of Q_n .

$$x \notin L \implies \Pr[Q_n(x) = 0] \leq \frac{1}{3} \quad (\text{soundness})$$

BQP contains classical computation

Every reversible classical computation is a quantum computation.

$$|x, y\rangle \mapsto |x, y \oplus f(x)\rangle$$

is a permutation, hence a unitary operation on computational basis states.

$$P \subseteq \text{BQP}.$$

BQP contains randomized computation

A quantum computer can generate a fair random bit:

$$|0\rangle \xrightarrow{H} \frac{|0\rangle + |1\rangle}{\sqrt{2}} \xrightarrow{\text{measure}} \begin{cases} 0 & \text{with probability } 1/2, \\ 1 & \text{with probability } 1/2. \end{cases}$$

Thus a quantum computer can simulate randomized classical computation.

$$\text{BPP} \subseteq \text{BQP}.$$

Error amplification

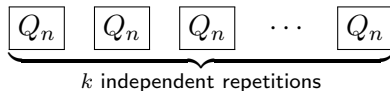
Suppose a BQP circuit is correct with probability at least $2/3$.

How can we boost the probability of being correct?

Error amplification

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How can we boost the probability of being correct?



Majority Vote



$$\Pr[\text{error}] \leq 2^{-\text{poly}(n)}.$$

BQP is robust

The definition of BQP does not depend on:

- the exact universal gate set,
- whether measurements occur during the computation or only at the end,
- $c = 2/3, s = 1/3$ or $c - s = \frac{1}{\text{poly}(|x|)}$

Independent Noise model: at each location i , independently, the implemented gate is

$$\tilde{U}_i = \begin{cases} U_i, & \text{with probability } 1 - p_i, \\ E_i, & \text{with probability } p_i, \end{cases} \quad p_i \leq p,$$

where E_i is some error. The overall computation is $\tilde{Q} = \tilde{U}_T \tilde{U}_{T-1} \cdots \tilde{U}_1$.

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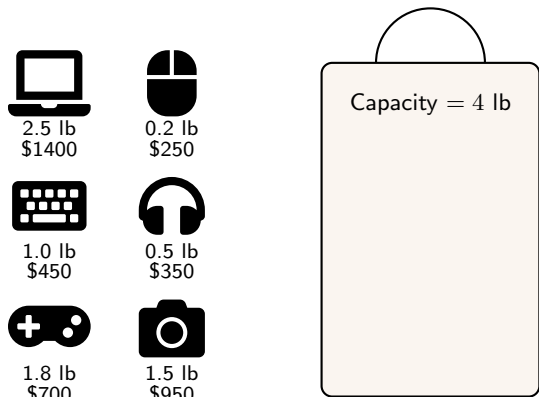
Theorem (Threshold (informal))

There exists a threshold $p_{\text{th}} > 0$ such that, whenever the error rate $p < p_{\text{th}}$, any quantum computation can be simulated to accuracy ε with only a polylogarithmic increase in circuit size.

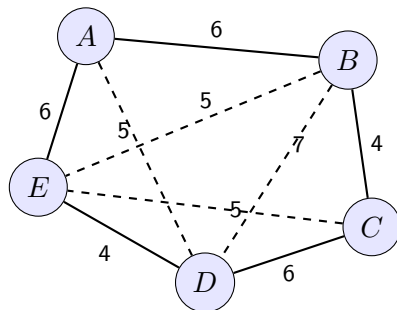
Problems Beyond Quantum Computers

Problems where quantum computers don't seem to help

Knapsack



Traveling salesman

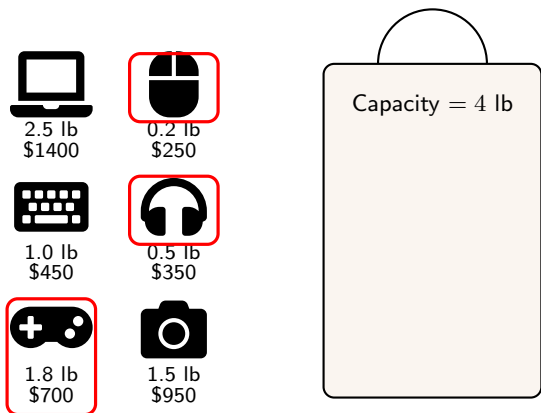


Is there a tour visiting every city once with 23km?

Can we pack items worth $\geq$ \$2300?

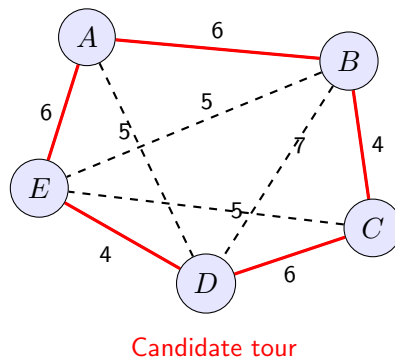
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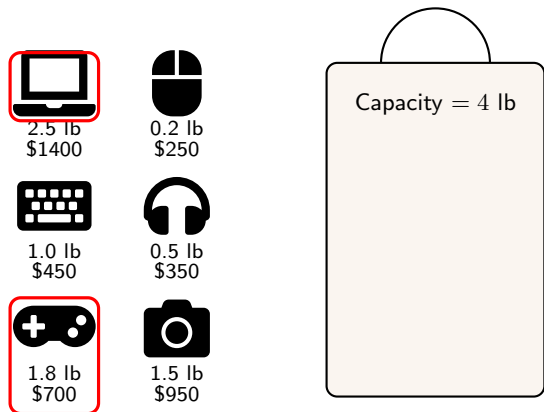
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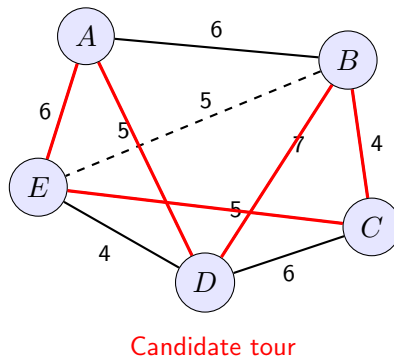
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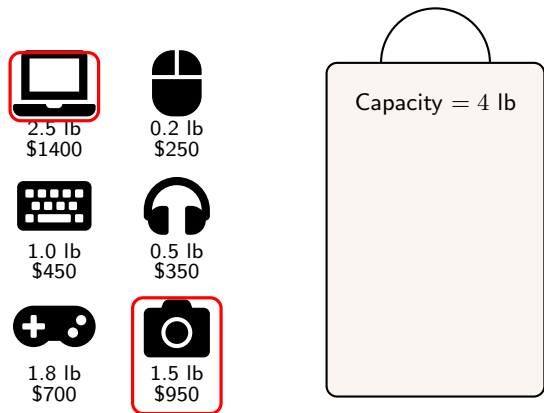
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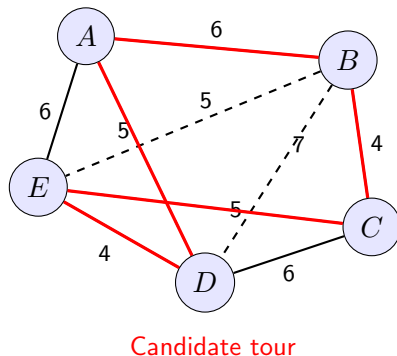
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Definition of NP (informal)

Traveling salesman

$$x \in \{0, 1\}^n$$



+ uniform circuits C_n
proof y

run $C_n(x, y)$
accept / reject

$L \in \text{NP}$ if there exists a uniform polynomial-time verifier C_n such that

$$x \in L \implies \exists y \in \{0, 1\}^{\text{poly}(n)} \quad C_n(x, y) = 1.$$

$$x \notin L \implies \forall y \in \{0, 1\}^{\text{poly}(n)} \quad C_n(x, y) = 0.$$

Adding Randomness: MA

Input arithmetic circuits for $A(x, y) \in \mathbb{F}[x_1, \dots, x_n, y_1, \dots, y_n]$ and $B(x) \in \mathbb{F}[x_1, \dots, x_n]$.

Yes instance: There exists $\mathbf{y}$ such that $A(x, \mathbf{y}) = B(x)$ for all x ?

How can a verifier solve this using a classical proof $\mathbf{y}$ + some randomness r ?

Adding Randomness: MA

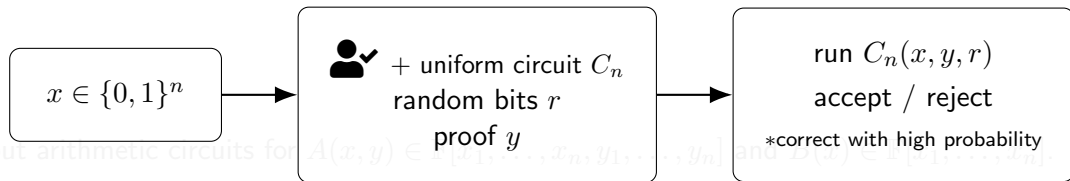
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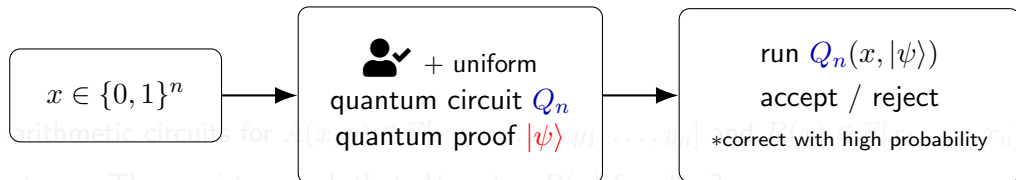
How can a verifier solve this using a classical proof $\mathbf{y}$ + some randomness r ?

- 1 Given $\mathbf{y}$ consider $A(x, \mathbf{y}) - B(x) \in \mathbb{F}[x_1, \dots, x_n]$
- 2 Check $A(x, \mathbf{y}) - B(x)$ on random points x as in the identity test protocol.

Definition of MA (informal)



Definition of QMA (informal)

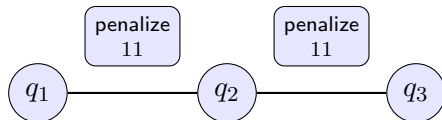


The k -Local Hamiltonian Problem

Example: A 2-local Hamiltonian

Consider the following 3-qubit Hamiltonian $H = |11\rangle\langle 11|_{12} + |11\rangle\langle 11|_{23}$.

The subscript 12 means the operator acts on qubits 1 and 2, and similarly for 23.

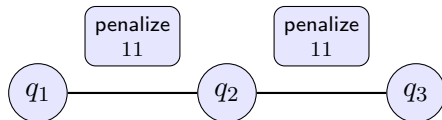


Does there exist a 3-qubit state $|\psi\rangle$ such that $\langle\psi|H|\psi\rangle = 0$?

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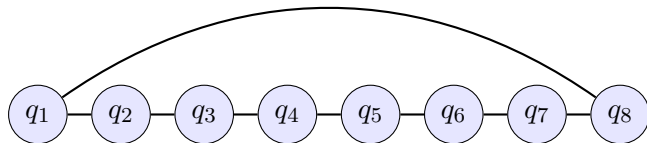
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We can take $\langle 000|H|000\rangle = 0$, since neither local term is violated. Therefore $\lambda_0(H) = 0$.

A slightly larger 2-local Hamiltonian

Consider the following 8-qubit Hamiltonian:

$$H = \sum_{i=1}^7 \left[\frac{1}{2}(I - Z_i Z_{i+1}) + \frac{1}{2}(I - X_i X_{i+1}) \right] + \frac{1}{2}(I - Z_1 Z_8) + \frac{1}{2}(I - X_1 X_8).$$

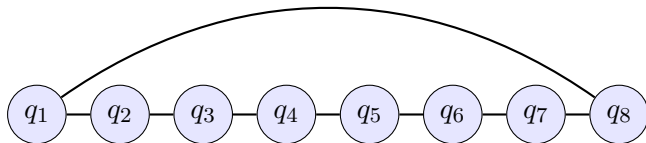


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Already for 8 qubits, the Hilbert space has dimension $2^8 = 256$.

The local terms are simple, but the global problem can be hard.

k -Local Hamiltonian

Recall: $H = H_1 + H_2 + \cdots + H_r$ is a k -local if $H_a = I^{\otimes i} \otimes K_a \otimes I^{\otimes (n-i-k)}$.

Let $\alpha, \beta : \mathbb{N} \rightarrow \mathbb{N}$ satisfying $\beta \geq \alpha$ and $\beta - \alpha = \frac{1}{\text{poly}(n)}$.

We want to turn this into a decision problem.

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k -local Hamiltonian Problem

Input: Hamiltonian $H = H_1 + H_2 + \cdots + H_r$ where,

- Each H_a acts on qubits $S_a \subseteq \{1, \dots, n\}$ with $|S_a| \leq k$,
- Local action is specified by $K_a \in L((\mathbb{C}^2)^{\otimes |S_a|})$. PSD operators satisfying $0 \leq K_a \leq I$.

YES : There exists an n -qubit state $|\psi\rangle$ such that $\langle \psi | H | \psi \rangle \leq \alpha$.

NO : Every n -qubit state $|\psi\rangle$ satisfies $\langle \psi | H | \psi \rangle \geq \beta$.

Promised only one of these is the case.

The k -local Hamiltonian problem is in QMA

Energy Test

Suppose $H = H_1 + H_2 + \cdots + H_r$ is a k -local Hamiltonian acting on n qubits,
With $0 \leq H_a \leq I$ and $H_a = I^{\otimes i} \otimes K_a \otimes I^{\otimes (n-i-k)}$, for some $K_a \in L((\mathbb{C}^2)^{\otimes k})$.

Energy Test: Given an n -qubit state $|\psi\rangle$ we can estimate $\langle \psi | H | \psi \rangle$.

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Energy Test: Given an n -qubit state $|\psi\rangle$ we can estimate $\langle\psi|H|\psi\rangle$.

- 1 Pick a term a uniformly at random and compute an eigendecomposition

$$K_a = \sum_j \lambda_j |v_j\rangle\langle v_j|.$$

- 2 Use a constant-size circuit U_a to implement

$$|v_j\rangle |0\rangle \mapsto |v_j\rangle \left(\sqrt{1-\lambda_j} |0\rangle + \sqrt{\lambda_j} |1\rangle \right) \quad (k \text{ system qubits} + \text{ancilla}).$$

- 3 Apply $I^{\otimes i} \otimes U_a \otimes I^{\otimes (n-i-k)}$ to $|\psi\rangle \otimes |0\rangle$, and measure ancilla. ($\Pr[1] = \langle\psi|H_a|\psi\rangle$).
- Overall $\Pr[\text{Accept}] = \Pr[0] = 1 - \frac{1}{r} \langle\psi|H|\psi\rangle$.

k-LH in QMA

Given $H = H_1 + H_2 + \cdots + H_r$

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Wait, doesn't QMA needs completeness $\frac{2}{3}$ and soundness $\frac{1}{3}$?

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Wait, doesn't QMA needs completeness $\frac{2}{3}$ and soundness $\frac{1}{3}$?

Theorem (Amplification)

If a QMA verifier has completeness c and soundness s with $c - s \geq 1/\text{poly}(n)$, then there is another verifier with completeness at least $2/3$ and soundness at most $1/3$.

(This can be amplified so that completeness is $1 - 2^{-p(n)}$ and soundness is $2^{-p(n)}$.)

Complete Problems

Reductions between problems

Sometimes we can solve one problem by transforming it into another.

Problem A: Given a graph G , does it contain an independent set of size k ?

Problem B: Given a graph H , does it contain a clique of size k ?

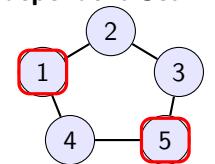
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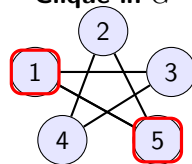
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Independent Set in G



independent set $\{1, 5\}$

Clique in $\overline{G}$



clique $\{1, 5\}$

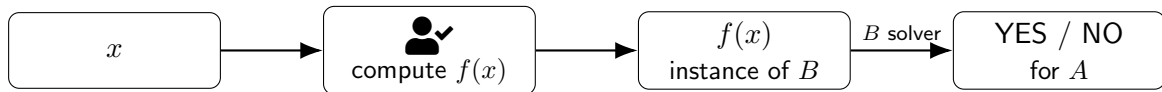
G has an independent set of size $k \iff \overline{G}$ has a clique of size k .

Complete problems

Reduction

A reduces to B , written $A \leq B$, if there is a polynomial-time computable function such that

$$x \in A \iff f(x) \in B.$$



A language B is called $\mathcal{C}$ -complete if $B \in \mathcal{C}$ and every language $A \in \mathcal{C}$ satisfies $A \leq B$.

Complete problems

Some examples from this course

Problem	In class?	Complete for?
Polynomial Identity Testing	BPP	-
Factoring ($n = pq$)	BQP, NP	Open question
Discrete Log (13^x)	BQP, NP	Open question
Knapsack	NP	NP
Travelling Salesman	NP	NP
$A(x, y) = B(x)$	MA	-
Local Hamiltonian	QMA	QMA

k-local Hamiltonian Problem is QMA-Complete

The Goal?

Recall: $L \in \text{QMA}$ if there exists a uniform quantum verifier Q_n such that

$$x \in L \implies \exists |\psi\rangle \quad \Pr[Q_n(|x\rangle |\psi\rangle |0 \cdots 0\rangle) \text{ accepts}] \geq 1 - 2^{-p(n)}.$$

$$x \notin L \implies \forall |\psi\rangle \quad \Pr[Q_n(|x\rangle |\psi\rangle |0 \cdots 0\rangle) \text{ accepts}] \leq 2^{-p(n)}.$$

Goal: $L \in \text{QMA}$, map $x \mapsto H_x$ with $x \in L \implies \lambda_0(H) \leq \alpha$ if $x \notin L \implies \lambda_0(H) \geq \beta$.

Candidate state

Construct k -local H whose ground state $|\Phi\rangle$ encodes the computation Q_n

Suppose $Q_n = U_T \cdots U_2 U_1$ acting on $|x\rangle|\phi\rangle|\bar{0}\rangle$. **We will omit ancilla register $|\bar{0}\rangle$.**

Each gate gives an intermediate state: $|x\rangle|\phi\rangle, U_1|x\rangle|\phi\rangle, U_2 U_1|x\rangle|\phi\rangle, \dots, U_T \cdots U_1|x\rangle|\phi\rangle$.

Step 1: Low energy states for H should look like a history of computation.

Target history state:

$$|\Phi\rangle = \frac{1}{\sqrt{T+1}} \sum_{t=0}^T \left(U_t \cdots U_1 |x\rangle|\phi\rangle \right) \otimes |t\rangle.$$

A Simpler problem

Target history state: $|\Phi\rangle = \frac{1}{\sqrt{T+1}} \sum_{t=0}^T \left(U_t \cdots U_1 |x\rangle |\phi\rangle \right) \otimes |t\rangle.$

Ignore the computation register for the moment. Find Hamiltonian with ground state:

$$\frac{1}{\sqrt{T+1}} \sum_{t=0}^T |t\rangle.$$

Define $H_{\text{clock}} = \sum_{t=0}^{T-1} H_t$, where $H_t = \frac{1}{2} \left(|t\rangle\langle t| + |t+1\rangle\langle t+1| - |t\rangle\langle t+1| - |t+1\rangle\langle t| \right).$

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Each term acts only on the span of $|t\rangle, |t+1\rangle$ and has matrix

$$\begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}, \quad (1, 1) \text{ has eigenvalue } 0, \quad (1, -1) \text{ has eigenvalue } 1.$$

The propagation matrix (simplified)

In the basis $|0\rangle, |1\rangle, \dots, |T\rangle$, the operator H_{clock} has matrix

$$H_{\text{clock}} = \frac{1}{2} \begin{bmatrix} 1 & -1 & & & & \\ -1 & 2 & -1 & & & \\ & -1 & 2 & -1 & & \\ & & \ddots & \ddots & \ddots & \\ & & & -1 & 2 & -1 \\ & & & & -1 & 1 \end{bmatrix}.$$

$\frac{1}{\sqrt{T+1}}(1, 1, \dots, 1) = \frac{1}{\sqrt{T+1}} \sum_{t=0}^T |t\rangle$ is a 0-eigenvector of H_{clock} .

The next smallest eigenvalue satisfies $\lambda_1(H_{\text{clock}}) \geq \frac{1}{2(T+1)^2}$.

Follows from the spectrum of the path graph random walk matrix.

The propagation matrix (simplified)

In the basis $|0\rangle, |1\rangle, \dots, |T\rangle$, the operator H_{clock} has matrix

This is $O(\log n)$ -local.

The clock states are encoded in binary using $s = \log_2(T + 1)$ qubits.

For example, $|3\rangle = |011\rangle$, $|4\rangle = |100\rangle$.

Thus the transition term

$$|t\rangle\langle t+1| = |t_1 t_2 \cdots t_s\rangle\langle t'_1 t'_2 \cdots t'_s|$$

acts on all $s = O(\log T) = O(\log n)$ qubits. (Since $T = \text{poly}(n)$.)

The next smallest eigenvalue satisfies $\lambda_1(H_{\text{clock}}) \leq \frac{1}{2(T+1)^2}$.

Follows from the spectrum of the path graph random walk matrix.

The propagation Hamiltonian

Target history state:

$$|\Phi\rangle = \frac{1}{\sqrt{T+1}} \sum_{t=0}^T \left(U_t \cdots U_1 |x\rangle |\phi\rangle \right) \otimes |t\rangle.$$

Define $H_{\text{prop}} = \sum_{t=1}^T H_t$, where H_t acts on the span of $|\psi_{t-1}\rangle|t-1\rangle$ and $U_t|\psi_{t-1}\rangle|t\rangle$.

$$H_t = \frac{1}{2} \begin{bmatrix} I & -U_t^\dagger \\ -U_t & I \end{bmatrix}.$$

Equivalently,

$$H_t = \frac{1}{2} \left(I \otimes |t\rangle\langle t| + I \otimes |t-1\rangle\langle t-1| - U_t \otimes |t\rangle\langle t-1| - U_t^\dagger \otimes |t-1\rangle\langle t| \right).$$

Invariant subspaces

Fix orthonormal bases $\{|a_j\rangle\}$ and $\{|\xi_k\rangle\}$ for the input and witness registers.

For each (j, k) , let $\mathcal{H}_{j,k}$ be the following subspace:

$$\mathcal{H}_{j,k} = \text{span} \{U_t \cdots U_1 |a_j\rangle |\xi_k\rangle \otimes |t\rangle : 0 \leq t \leq T\}.$$

The subspaces $\mathcal{H}_{j,k}$ are mutually orthogonal, and invariant ($H_{\text{prop}} \cdot \mathcal{H}_{j,k} \subseteq \mathcal{H}_{j,k}$.)

Thus H_{prop} decomposes as a direct sum of identical $(T+1) \times (T+1)$ propagation matrices.

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Thus H_{prop} decomposes into propagation matrices.

	$\mathcal{H}_{0,0}$	$\mathcal{H}_{0,1}$	$\mathcal{H}_{j,k}$	...
$\mathcal{H}_{0,0}$	$A_{0,0}$	0	...	0
$\mathcal{H}_{0,1}$	0	$A_{0,1}$	...	0
$\mathcal{H}_{j,k}$	0	0	$A_{j,k}$	0
...	...	...	...	$A_{M,N}$

$$A_{j,k} = H_{\text{prop}}|_{\mathcal{H}_{j,k}}$$

Basis of $\mathcal{H}_{j,k}$: $|a_j\rangle|\xi_k\rangle|0\rangle, U_1|a_j\rangle|\xi_k\rangle|1\rangle, \dots$

Spectrum of H_{prop}

Let $A_{j,k} := H_{\text{prop}}|_{\mathcal{H}_{j,k}}$. In the basis $U_t \cdots U_1 |a_j\rangle |\xi_k\rangle \otimes |t\rangle$ for $0 \leq t \leq T$,

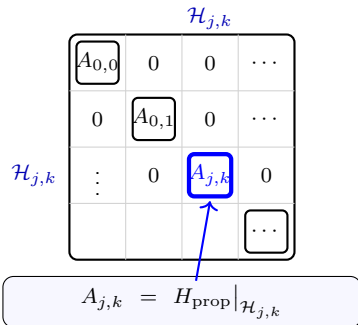
$$A_{j,k} = \frac{1}{2} \begin{bmatrix} 1 & -1 & & & \\ -1 & 2 & -1 & & \\ & -1 & 2 & -1 & \\ & & \ddots & \ddots & \ddots \\ & & & -1 & 2 & -1 \\ & & & & -1 & 1 \end{bmatrix}.$$

Thus $\ker(H_{\text{prop}})$ is spanned by

$$|\Psi_{j,k}\rangle = \frac{1}{\sqrt{T+1}} \sum_{t=0}^T U_t \cdots U_1 |a_j\rangle |\xi_k\rangle \otimes |t\rangle.$$

Since $H_{\text{prop}} \cong \bigoplus_{j,k} A_{j,k}$,

Block structure



Properties of H_{prop}

Candidate history state: $|\Phi\rangle = \frac{1}{\sqrt{T+1}} \sum_{t=0}^T (U_t \cdots U_1 |x\rangle |\phi\rangle) \otimes |t\rangle.$

Since $A_{j,k} = H_{\text{prop}}|_{\mathcal{H}_{j,k}}$, every 0-energy state is a superposition of valid histories:

$$|\Psi\rangle := \frac{1}{\sqrt{T+1}} \sum_{t=0}^T (U_t \cdots U_1 |z\rangle |\psi\rangle) \otimes |t\rangle$$

for some input $|z\rangle$ and witness $|\psi\rangle$.

Moreover, $\lambda_1(H_{\text{prop}}) = \lambda_1(A_{j,k}) \geq \frac{1}{2(T+1)^2}.$

Recap

Fix $L \in \text{QMA}$ with quantum verifier $Q_n = U_T \cdots U_1$.

Goal: given input x , construct a local Hamiltonian $x \mapsto H_x$ such that

$$\text{Yes} \quad \exists |\phi\rangle \quad \Pr[Q_n(|x\rangle |\phi\rangle) = 1] \geq 1 - 2^{-\text{poly}(n)} \implies \lambda_0(H_x) \leq \alpha,$$

$$\text{No} \quad \forall |\phi\rangle \quad \Pr[Q_n(|x\rangle |\phi\rangle) = 1] \leq 2^{-\text{poly}(n)} \implies \lambda_0(H_x) \geq \beta.$$

Step 1: Low energy states for H look like a history of computation. ✓

What we have so far

$$H = H_{\text{prop}}.$$

H_{prop} has ground space spanned by the computational history states

$$\frac{1}{\sqrt{T+1}} \sum_{t=0}^T \left(U_t \cdots U_1 |z\rangle |\psi\rangle \right) \otimes |t\rangle.$$

$$\lambda_1(H_{\text{prop}}) \geq \frac{1}{2(T+1)^2}.$$

Step 2: Low energy states for H should be history states with $z = x$.

Initialization check

H_{prop} only verifies that the computation evolves consistently.

$$|\Psi\rangle = \frac{1}{\sqrt{T+1}} \sum_{t=0}^T \left(U_t \cdots U_1 |z\rangle |\psi\rangle \right) \otimes |t\rangle$$

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We could have $z \neq x$.

We add another term $H = H_{\text{in}} + H_{\text{prop}}$

Write $x = (x_1, \dots, x_n)$ and let $\overline{x_i} = 1 - x_i$. Define

$$H_{\text{in}} = \sum_{i=1}^n |\overline{x_i}\rangle \langle \overline{x_i}|_i \otimes |0\rangle \langle 0|_{\text{clock}}.$$

If a valid history state begins with $z \neq x$, then some bit differs, so $\langle \Psi | H_{\text{in}} | \Psi \rangle \geq \frac{1}{T+1}$.

Recap

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Step 1: Low energy states for H look like a history of computation. ✓

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What we have so far

$H = H_{\text{in}} + H_{\text{prop}}$. Let $|v\rangle$ be an arbitrary state.

- If $|v\rangle$ is not (close to) a history state, then $\langle v | H_{\text{prop}} | v \rangle$ is large.
(Since the 0-eigenspace of H_{prop} is spanned by history states and $\lambda_1(H_{\text{prop}}) \geq \frac{1}{2(T+1)^2}$.)

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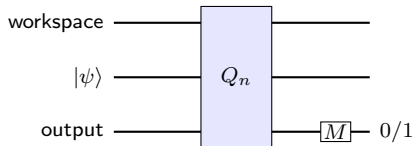
Any low-energy state must be (approximately) a valid history state whose computation begins with the input x .

Step 3: Low energy states for H must be history states with $z = x$ and a valid proof.

Checking acceptance at the final time step

Previously we wrote $\Pr[Q_n(|x\rangle |\psi\rangle) = 1]$.

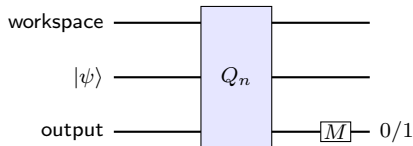
More explicitly, the verifier produces the state $U_T \cdots U_1 |x\rangle |\psi\rangle |0 \cdots 0\rangle$, then measures a designated output qubit.



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More explicitly, the verifier produces the state $U_T \cdots U_1 |x\rangle |\psi\rangle |0 \cdots 0\rangle$, then measures a designated output qubit.



Define

$$H_{\text{out}} = |0\rangle\langle 0|_{\text{out}} \otimes |T\rangle\langle T|_{\text{clock}}.$$

If $|\psi\rangle$ is rejected with probability p , then $\langle \Psi | H_{\text{out}} | \Psi \rangle \geq \frac{p}{T+1}$.

Recap

Fix $L \in \text{QMA}$ with quantum verifier $Q_n = U_T \cdots U_1$.

Goal: given input x , construct a local Hamiltonian $x \mapsto H_x$ such that

$$\text{Yes} \quad \exists |\phi\rangle \quad \Pr[Q_n(|x\rangle |\phi\rangle) = 1] \geq 1 - 2^{-\text{poly}(n)} \implies \lambda_0(H_x) \leq \alpha,$$

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Step 1: Low energy states for H look like a history of computation. ✓

Step 2: Low energy states for H should be history states with $z = x$. ✓

Step 3: Low energy states for H must be history states with $z = x$ and a valid proof. ✓

Completeness

Suppose $x \in L$. Then there exists $|\phi\rangle$ such that $\Pr[Q_n(|x\rangle |\phi\rangle) = 1] \geq 1 - 2^{-\text{poly}(n)}$.

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$$\text{Let } |\Phi\rangle = \frac{1}{\sqrt{T+1}} \sum_{t=0}^T \left(U_t \cdots U_1 |x\rangle |\phi\rangle \right) \otimes |t\rangle.$$

Then for $H = H_{\text{prop}} + H_{\text{in}} + H_{\text{out}}$

$$\langle \Phi | H_{\text{prop}} | \Phi \rangle = 0, \quad \langle \Phi | H_{\text{in}} | \Phi \rangle = 0,$$

since $|\Phi\rangle$ is a valid history beginning with input x .

Also,

$$\langle \Psi | H_{\text{out}} | \Psi \rangle = \frac{1}{T+1} \Pr[Q_n(|x\rangle |\psi\rangle) = 0] \leq \frac{2^{-\text{poly}(n)}}{T+1}.$$

Hence $\lambda_0(H) \leq 2^{-\text{poly}(n)}$.

Soundness: valid history states

Now suppose $x \notin L$. Then every $|\psi\rangle$ satisfies $\Pr[Q_n(|x\rangle|\psi\rangle) = 0] \geq 1 - 2^{-\text{poly}(n)}$.

$$\text{Recall } \ker(H_{\text{prop}}) = \text{span} \left\{ \frac{1}{\sqrt{T+1}} \sum_{t=0}^T \left(U_t \cdots U_1 |z\rangle |\psi\rangle \right) \otimes |t\rangle \right\}.$$

$$\text{If } z \neq x \text{ then } \langle \Phi | H_{\text{in}} | \Phi \rangle \geq \frac{1}{T+1}.$$

Moreover, for valid histories beginning with $z = x$,

$$\langle \Phi | H_{\text{out}} | \Phi \rangle = \frac{1}{T+1} \Pr[Q_n(|x\rangle|\psi\rangle) = 0] \geq \frac{1 - 2^{-\text{poly}(n)}}{T+1}.$$

Hence every state $|\Phi\rangle \in \ker(H_{\text{prop}})$ satisfies $\langle \Phi | (H_{\text{in}} + H_{\text{out}}) | \Phi \rangle \geq \frac{1 - 2^{-\text{poly}(n)}}{(T+1)}$.

Soundness: arbitrary states (Informal)

Recall that $\lambda_1(H_{\text{prop}}) \geq \frac{1}{2(T+1)^2}$.

Now let $|\eta\rangle$ be an arbitrary state and decompose

$$|\eta\rangle = |\eta_0\rangle + |\eta_\perp\rangle, \quad |\eta_0\rangle \in \ker(H_{\text{prop}}), \quad |\eta_\perp\rangle \perp \ker(H_{\text{prop}}).$$

Since every nonzero eigenvalue of H_{prop} is at least $\frac{1}{2(T+1)^2}$,

$$\langle \eta | H_{\text{prop}} | \eta \rangle \geq \frac{1}{2(T+1)^2} \|\eta_\perp\|^2.$$

On the other hand, from the previous slide every valid history state $|\Phi\rangle \in \ker(H_{\text{prop}})$ satisfies

$$\langle \Phi | (H_{\text{in}} + H_{\text{out}}) | \Phi \rangle \geq \frac{1 - 2^{-\text{poly}(n)}}{T+1}.$$

Thus:

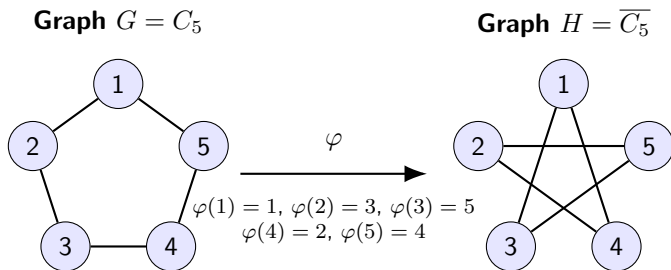
- either $|\eta\rangle$ is far from $\ker(H_{\text{prop}})$, and H_{prop} gives energy,
- or $|\eta\rangle$ is close to $\ker(H_{\text{prop}})$, and then $H_{\text{in}} + H_{\text{out}}$ gives energy.

Interactive Proof Systems

Graph Isomorphism

G and H are *isomorphic* if there exists an adjacency preserving bijection $\phi : G \rightarrow H$.

Isomorphism Problem: Given graphs G and H , decide if G is isomorphic to H .

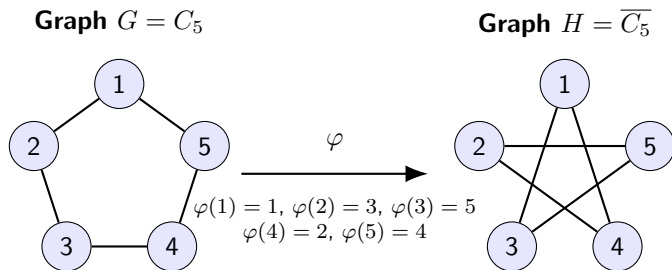


Which complexity class we have seen can verify a proposed isomorphism?

Graph Isomorphism

G and H are *isomorphic* if there exists an adjacency preserving bijection $\phi : G \rightarrow H$.

Isomorphism Problem: Given graphs G and H , decide if G is isomorphic to H .



Which complexity class we have seen can verify a proposed isomorphism?

GRAPH ISOMORPHISM $\in$ NP

Graph Non-Isomorphism

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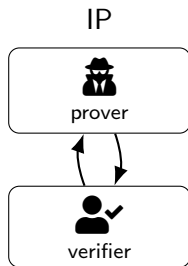
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IP is the set of problems for which there exists an interactive protocol such that

$$x \in L \implies \Pr[\text{accept}] \geq \frac{2}{3}, \quad x \notin L \implies \Pr[\text{accept}] \leq \frac{1}{3}.$$

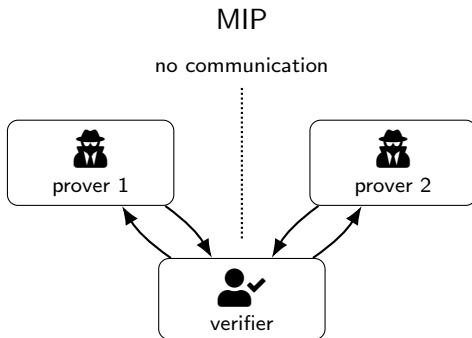
Multi-Prover Interactive Proofs



One prover



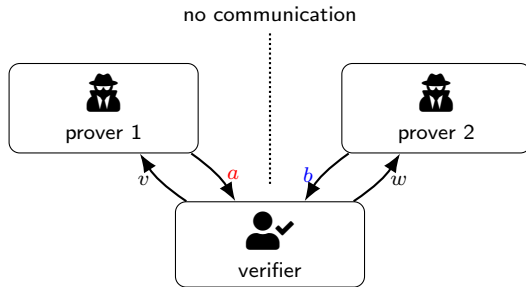
Multiple non-communicating provers



MIP $\Leftrightarrow$ Nonlocal Games

A Two-Prover Graph Coloring Game

How can two provers convince a verifier that G is 3-colorable?



Verifier's test

- 1 Sample an edge $v - w$ and send v to prover 1 and w to prover 2.
- 2 Receive colors $a, b \in \{\text{red}, \text{blue}, \text{green}\}$.
- 3 Accept if $a \neq b$.

G is 3-colorable $\iff$ the provers can always win this game.

A Two-Prover Graph Coloring Game

What if the provers are allowed to share an entangled state?

$$|\psi\rangle \in \mathbb{C}^d \otimes \mathbb{C}^d$$

Important: Entanglement does *not* allow provers to communicate.

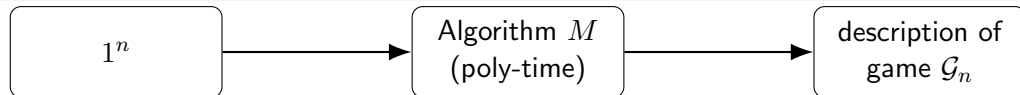


There exist graphs that are **not** 3-colorable, yet entangled provers can win the 3-coloring game with probability 1.

Uniform Games

Uniformity

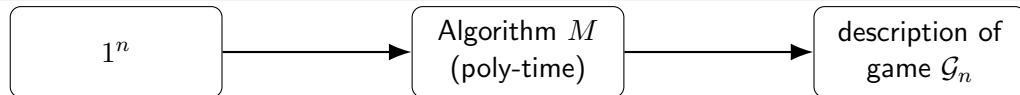
A family of games $\{\mathcal{G}_n\}_{n \geq 1}$ is *uniform* if there exists a polynomial p and an algorithm M which, on input 1^n , outputs a description of the circuit $\mathcal{G}_n$ in at most $p(n)$ steps.



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$$L \in \text{MIP}^*$$

if there exists a uniform family of games $\mathcal{G}_n$ such that

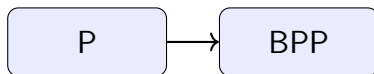
$$x \in L \implies \omega^*(\mathcal{G}_n) \geq \frac{2}{3} \quad (\text{completeness})$$

$$x \notin L \implies \omega^*(\mathcal{G}_n) \leq \frac{1}{3} \quad (\text{soundness})$$

Where do these classes fit?

P

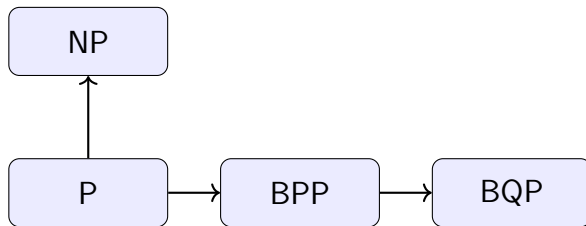
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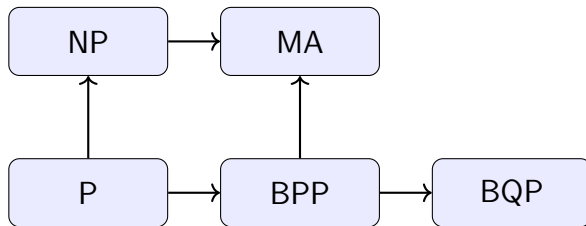
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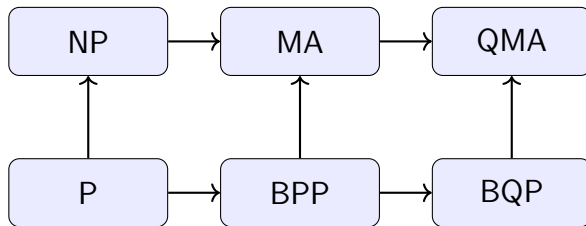
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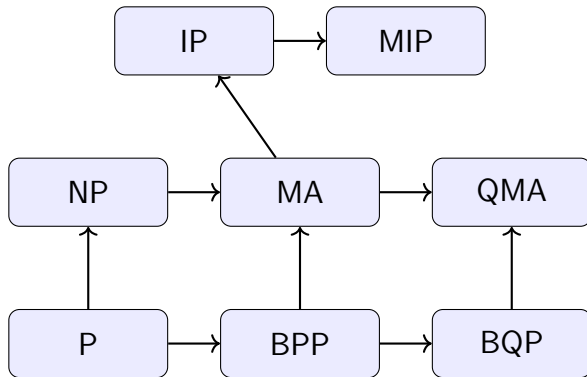
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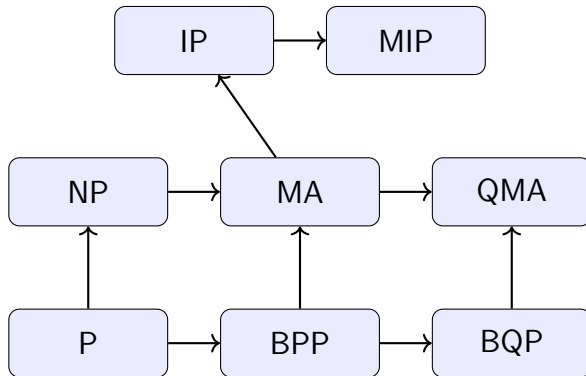


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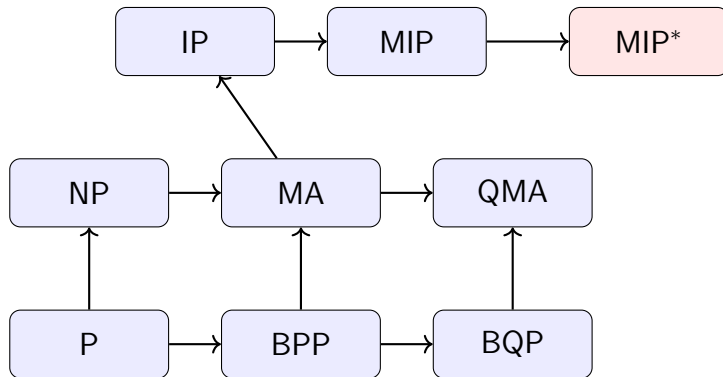
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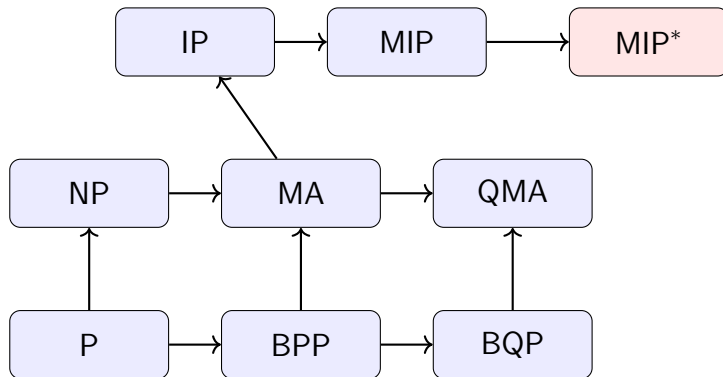


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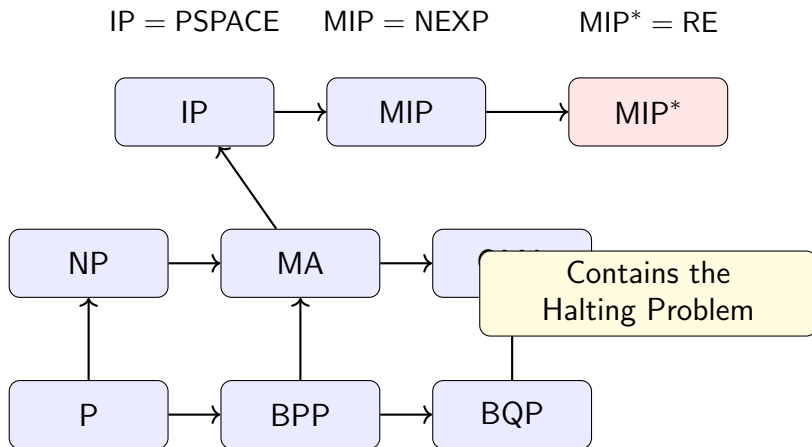
$IP = PSPACE$

$MIP = NEXP$

$MIP^* = RE$



Where do these classes fit?



Self-Testing

CHSH: inputs: $x, y \in \{0, 1\}$ outputs: $a, b \in \{0, 1\}$ win iff: $a + b = xy \pmod{2}$

Alice and Bob can win with probability $\geq \cos^2\left(\frac{\pi}{8}\right) \approx 0.854$,

using state $|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$, and observables $(X, Z$ for Alice and $\frac{X+Z}{\sqrt{2}}, \frac{X-Z}{\sqrt{2}}$ for Bob),

Self-testing for CHSH (informal)

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- ③ If the win rate is $0.854 - \epsilon$ the state and observables are $O(\sqrt{\epsilon})$ -close to the ideal. (2013)

Our next goals:

- 1 Prove Tsirelson's bound $\omega^*(\text{CHSH}) \leq \cos^2\left(\frac{\pi}{8}\right)$.
- 2 Explain the high-level idea of self-testing.

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.....

An entangled strategy consists of

State: $|\psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$ **PVMs:** $\{P_x^a\} \subseteq B(\mathcal{H}_A), \{Q_y^b\} \subseteq B(\mathcal{H}_B).$

$$\Pr[\text{win}] = \frac{1}{4} \sum_{\substack{x,y,a,b \\ a+b=xy}} \langle \psi | P_x^a \otimes Q_y^b | \psi \rangle.$$

We define the following observables (order 2 unitaries)

Alice's observables $A_x := P_x^0 - P_x^1$ **Bob's observables** $B_y := Q_y^0 - Q_y^1$

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$$2\sqrt{2} - 4\beta = \frac{\sqrt{2}}{4} \langle \psi | \left(A_0 \otimes I - \frac{I \otimes B_0 + I \otimes B_1}{\sqrt{2}} \right)^2 + \left(A_1 \otimes I - \frac{I \otimes B_0 - I \otimes B_1}{\sqrt{2}} \right)^2 | \psi \rangle$$

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$$\implies 2\sqrt{2} - 4\beta \geq 0$$

$$\beta \leq \frac{1}{\sqrt{2}}$$

$$\Pr(\text{win}) \leq \frac{1}{2\sqrt{2}} + \frac{1}{2}.$$

Anti-Commutation Relations

Recall: In the optimal strategy Alice uses

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

What is $XZ + ZX = ?$

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$$XZ = -ZX.$$

This is a powerful relation!

Lemma

If $A, B \in M_2(\mathbb{C})$ are self-adjoint unitaries and $AB = -BA$, then

$$UAU^\dagger = Z, \quad UBU^\dagger = X \quad \text{for some } U.$$

Proof sketch:

$$A = A^\dagger, A^2 = I \implies A \sim \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = Z.$$

Writing

$$B = \begin{pmatrix} a & c \\ \bar{c} & d \end{pmatrix},$$

the relation $ZB = -BZ$ forces $a = d = 0$, $|c| = 1$.

Anti-commutation for Alice

Suppose players are using an optimal strategy in $\mathbb{C}^2 \otimes \mathbb{C}^2$

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(Simplifying assumption:) $|\psi\rangle$ has full Schmidt rank.

$$\implies A_0^T = \frac{B_0 + B_1}{\sqrt{2}} \quad A_1^T = \frac{B_0 - B_1}{\sqrt{2}} \implies A_0 A_1 = -A_1 A_0.$$

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$$\implies A_0 \sim Z \text{ and } A_1 \sim X, \quad \sqrt{2} B_0 \sim Z + X \text{ and } \sqrt{2} B_1 \sim Z - X.$$

Eigenvalue analysis of $A_0 \otimes B_0 + A_1 \otimes B_0 + A_0 \otimes B_1 - A_1 \otimes B_1 \implies |\psi\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle).$

High-level idea of self-testing

Self-testing lets the verifier certify what state is present and what measurements are being performed.

$$\text{Example : } H = \frac{1}{2}(I - X_1X_2) + \frac{1}{2}(I - Z_2Z_3) + \frac{1}{2}(I - X_1Z_3).$$

A suitable non-local game can force the provers to approximately measure the observables

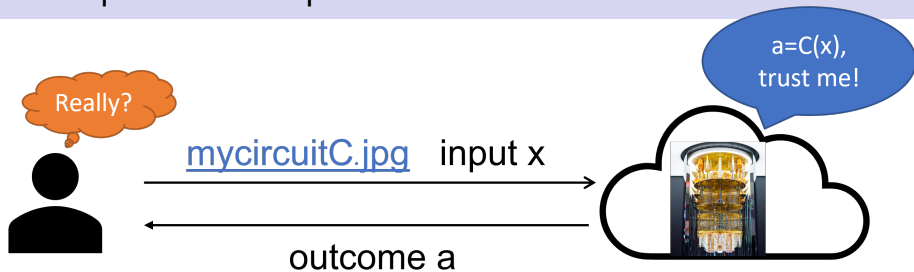
$$X_1X_2, \quad Z_2Z_3, \quad X_1Z_3 \quad \text{on a shared state } |\psi\rangle.$$

Verifier certifies local Hamiltonian measurements $\implies$ Verifier certifies a QMA witness.

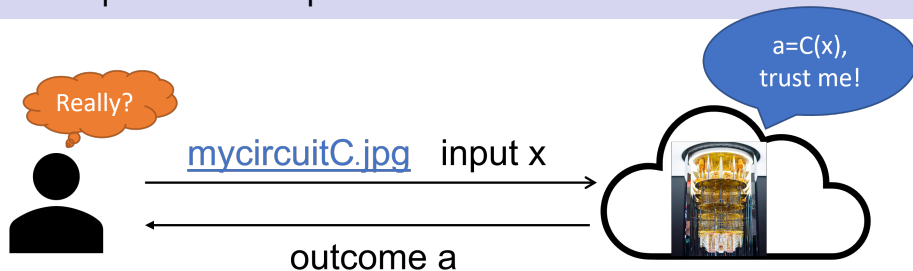
$$\text{QMA} \subseteq \text{MIP}^*.$$

Applications to Verifiable Delegation

Delegation of quantum computations



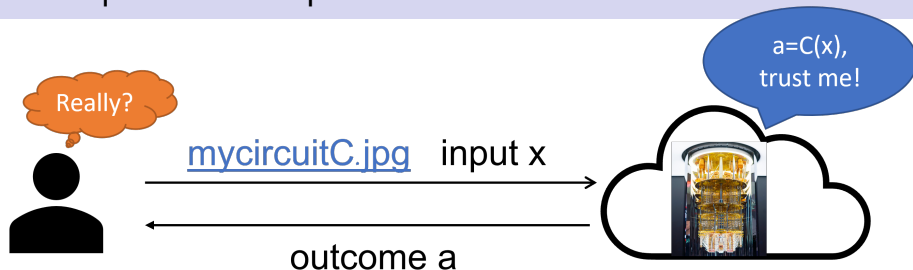
Delegation of quantum computations



Client: verifiable delegation

- I want to be convinced of the correctness,
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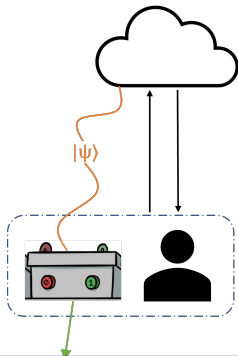
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Server: zero-knowledge proof

- We need to convince our clients that we are honest,
- but we don't want to reveal any inner-workings

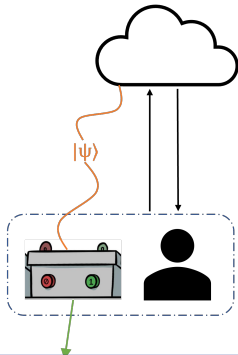
2 Device Delegation



A measurement device

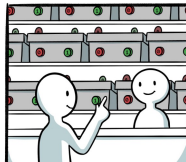
A small untrusted measurement device supplied by the server.

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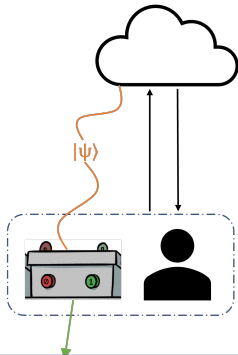


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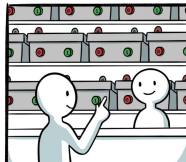


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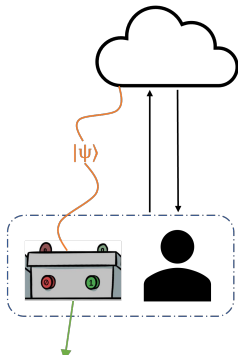
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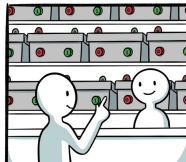
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 $\mathcal{C} \rightarrow H: \mathcal{C} \text{ accepts} \Leftrightarrow \lambda_0(H) \text{ is small}$

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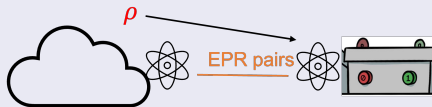


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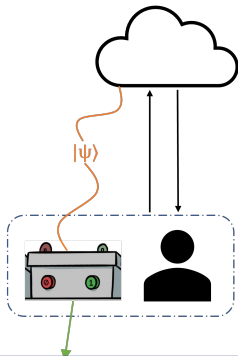
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- 1 Circuit-to-Hamiltonian construction $\mathcal{C} \rightarrow H$: $\mathcal{C}$ accepts $\Leftrightarrow \lambda_0(H)$ is small
- 2 Teleport the ground state ρ of H

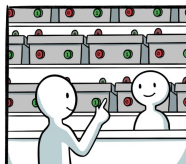


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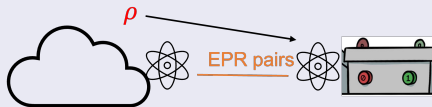


A measurement device

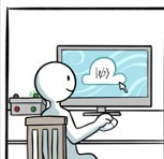
A small untrusted measurement device supplied by the server.



- 1 Circuit-to-Hamiltonian construction $\mathcal{C} \rightarrow H$: $\mathcal{C}$ accepts $\Leftrightarrow \lambda_0(H)$ is small
- 2 Teleport the ground state ρ of H



- 3 Estimates the ground energy of H



- Accepts if $\lambda_0(H)$ is low
- Rejects if $\lambda_0(H)$ is high

How Powerful is MIP^* ?

The Modern EPR paradox

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$$A_x \otimes B_y, |\psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B.$$

- 2 Infinite dimensional, based on relativistic quantum mechanics.

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[Ji, Natarajan, Vidick, Wright, Yuen (2020)] No.

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- 4 Group C^* -algebras

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Theorem (Navascués, Pironio, Acín (2008))

If the answer to Connes' or Tsirelson's problem is yes then there exists an algorithm that computes the quantum win rate of a game.

Theorem Ji, Natarajan, Vidick, Wright, Yuen (2020):
There exists a non-local game for the halting problem.

